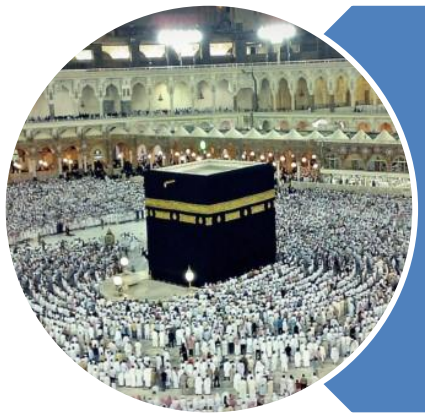
Data Structures - I

CPCS 204

رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي وَاحْلُلْ عُقْدَةً مِنْ لِسَانِي يَفْقَهُوا قَوْلِي



O my Lord! Expand for me my chest [with assurance] and ease for me my task and untie the knot from my tongue that they may understand my speech. **(Quran 20 : 25-28)**

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CPCS 204

Heap

Introduction

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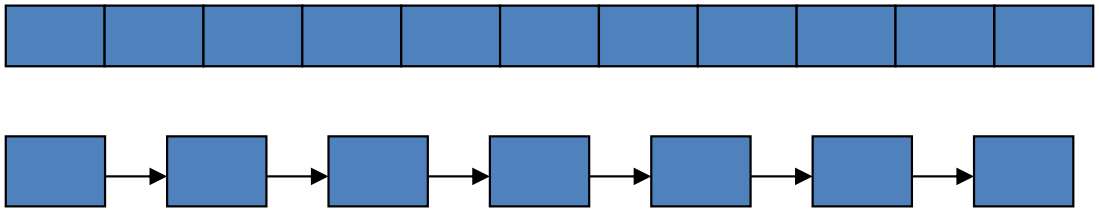
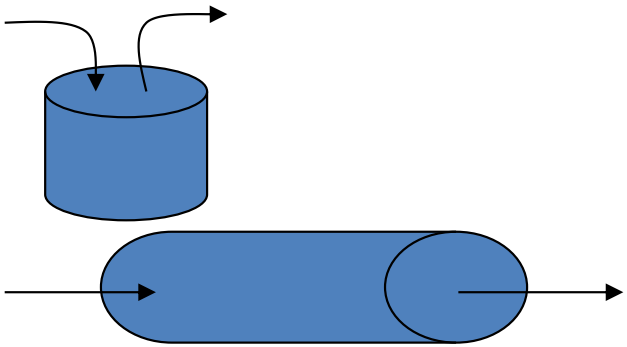
Trees - Introduction

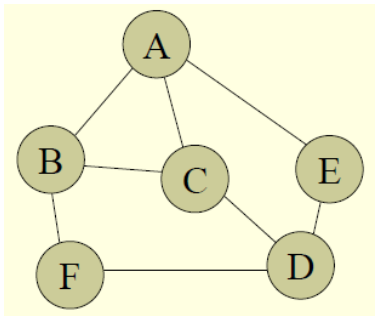
- Arrays
- Linked List

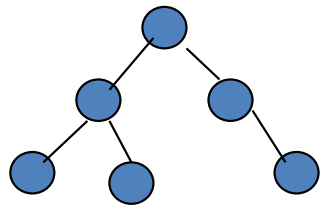


- Stacks
- Queues



- **Tree**
- Graph





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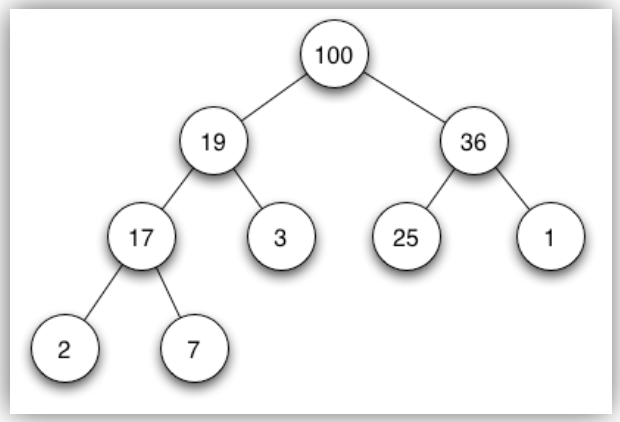
Heap - Introduction

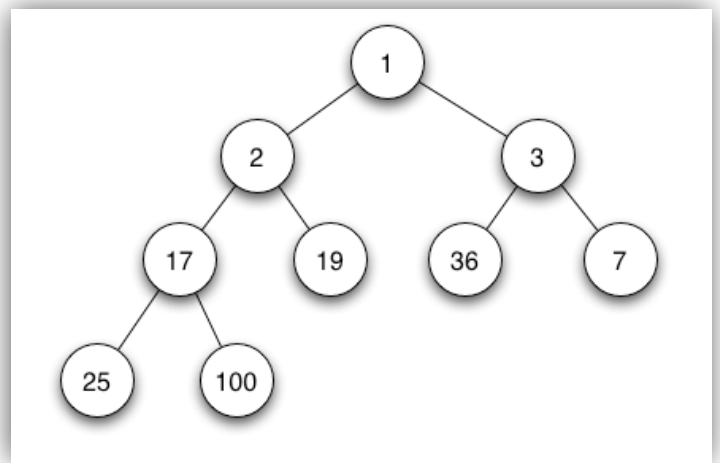
- Definition
  - A heap is an Abstract Data Type, is a variant of tree data structure



- Heap Properties
  - Order / heap Property
    - Children **Values are arranged** with respect to root
  - Shape Property
    - **Complete** Binary Tree

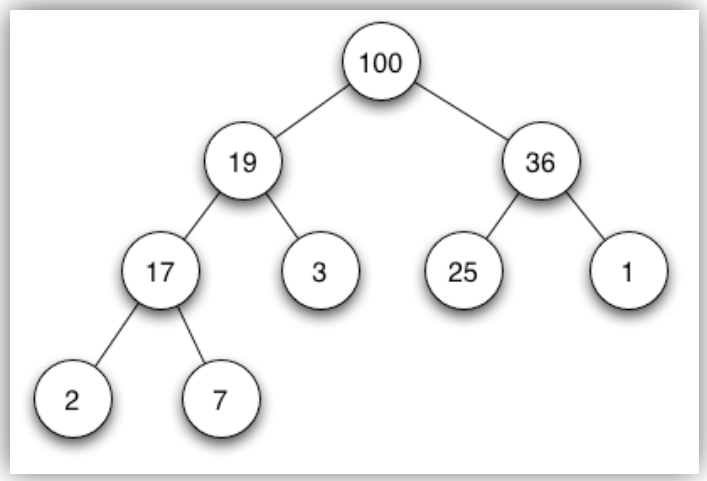
Heap - Types

- Min – Heap
  - If all values stored in the **root** of a given tree is **smaller** than the value stored in the subtrees (**Order / heap Property**)
  - It is Complete Binary Tree (**Shape Property**)



Heap - Types

- Max – Heap
  - If all values stored in the **root** of a given tree is **greater or equal** to the value stored in the subtrees (**Order / heap Property**)
  - It is Complete Binary Tree (**Shape Property**)



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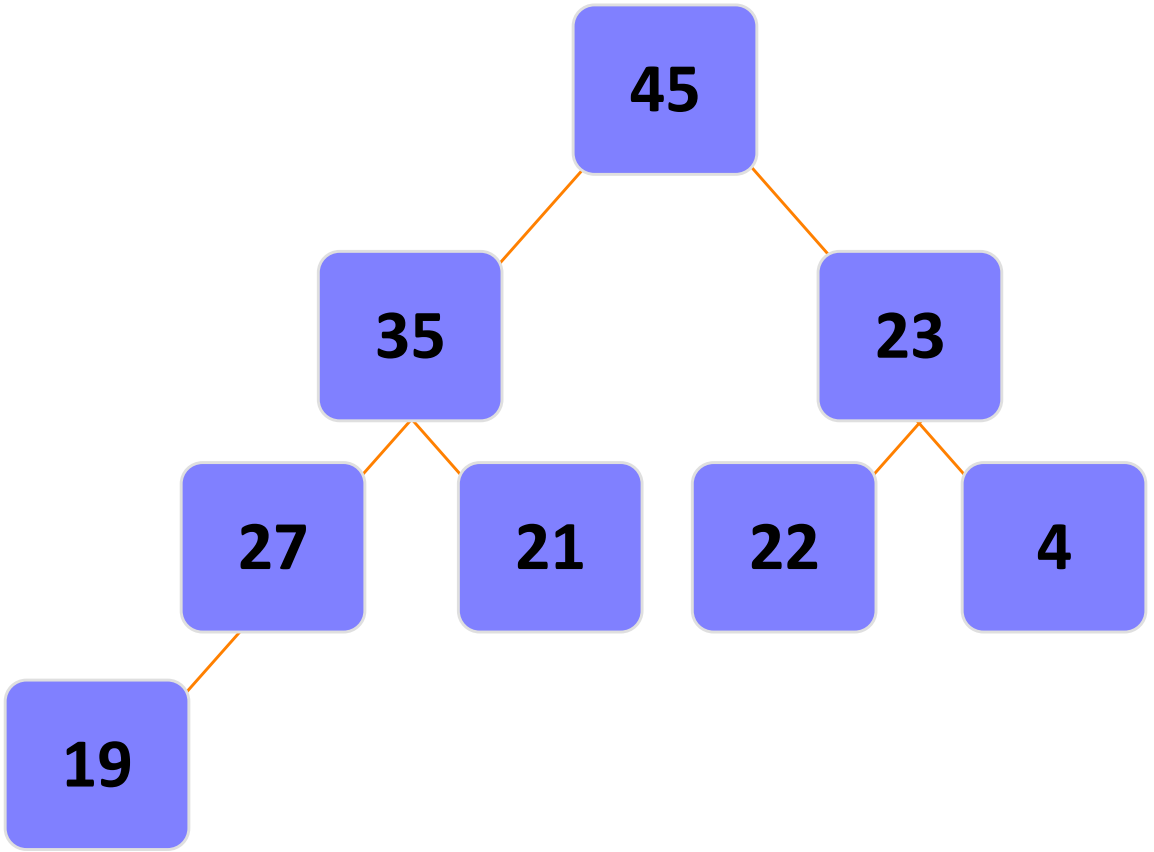
Heap – **Building Up**

- This is called
  - **Max - Heap**



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Heap – **Building Up**

- New nodes are always added at the lowest level
  - And are inserted from left to right
- There is no particular relationship among the data items in nodes on any given level
  - Even if the nodes have the same parent
  - Example: the right node does not necessarily have to be larger than the left node (as in BSTs)
- The only ordering property for heaps is the one already defined
  - Root of any given subtree is either largest or smallest element in that tree…either a max-heap or a min-heap

Heap – **Building Up**

- The tree never becomes unbalanced
- A heap is not a sorted structure
  - But it can be regarded as partially ordered
    - Since the minimum / maximum value is always at the root
- A given set of data can be formed into many different heaps
  - Depending on the order in which the data arrives

CPCS 204

Heap

Operations

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Heap - **Operations**

- Adding Node
  - Insertion
- Deleting Node
  - Deletion
- Building Heap
  - Creation

Heap Operations

Insertion

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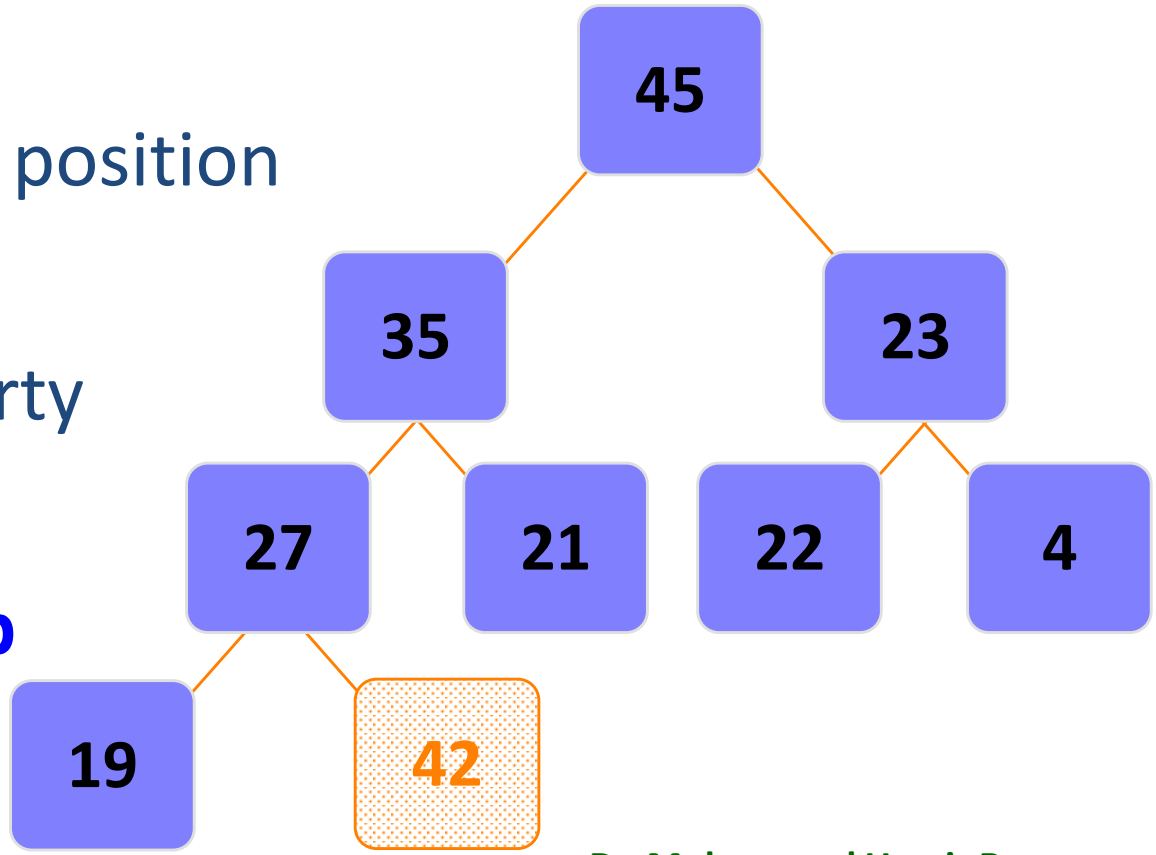
Operation - **Insertion**

- Assume the existence of a current heap
- Remember:
  - The binary heap MUST follow the Shape property
    - The tree must be balanced
- Insertions will be made in the **next available spot**
  - Meaning, at the last level
  - and at the next spot, going from left to right
- But what will most likely happen when you do this?
  - **The Heap / Order property will NOT be maintained**

Operation - Insertion

- Given Max-Heap
- Add "42"
  - Insert at last position
- It violates
  - Order property
- Fix it
  - **Percolate Up**



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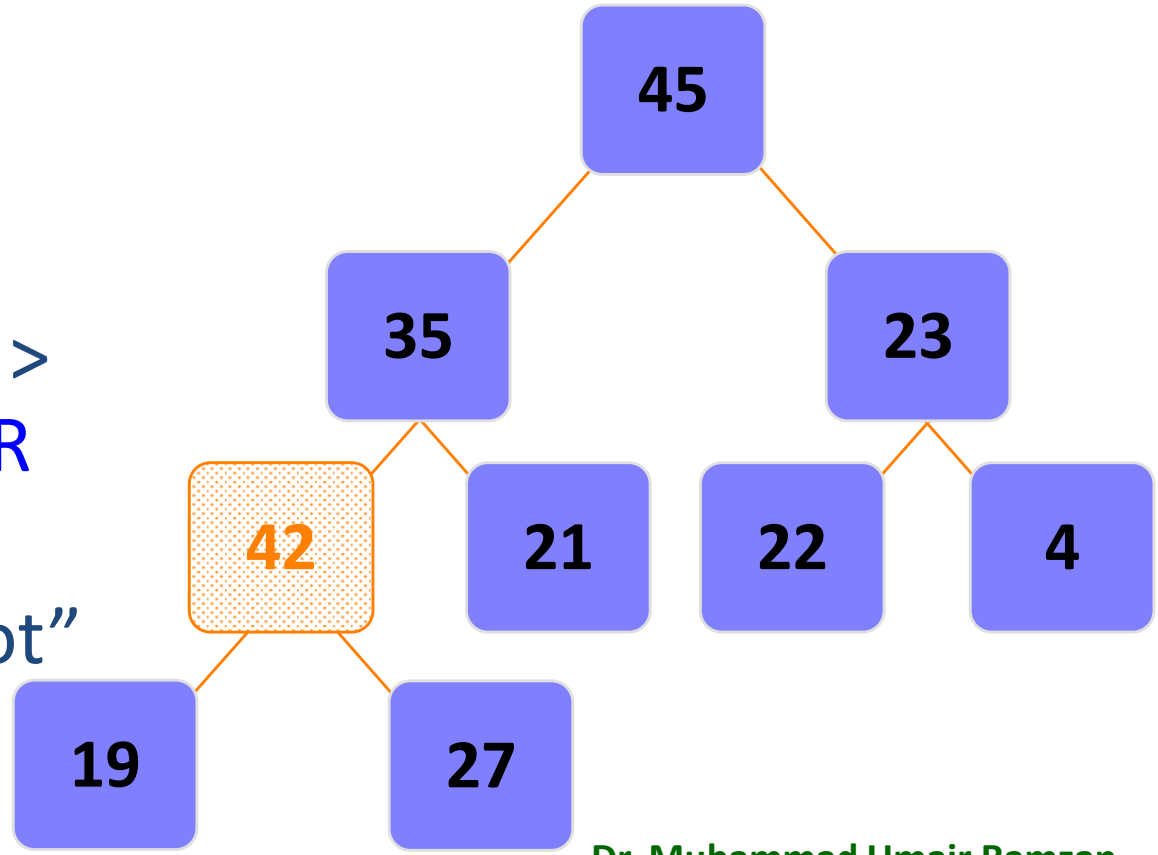
Insertion – Percolate Up

- If new node > parent node
  - Swap them
- Continue till
  - Parent node > new node OR
  - New node become "root"



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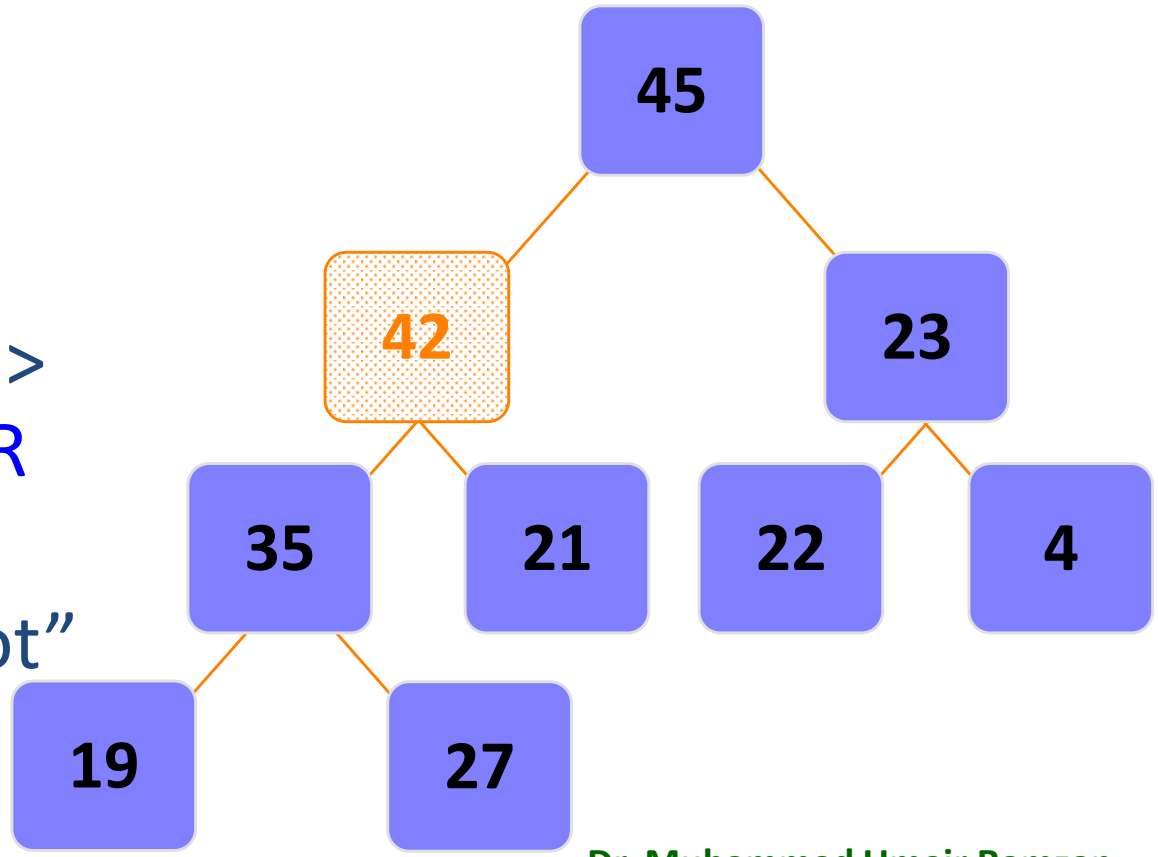

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Insertion – **Analysis**

- What is the Big-O running time of insertion into a heap?
- The actual insertion is simply
  - O(1)
- Percolate Up
  - Height of tree
  - O(logn)
- Overall **O(logn)**

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Heap Operations

Deletion

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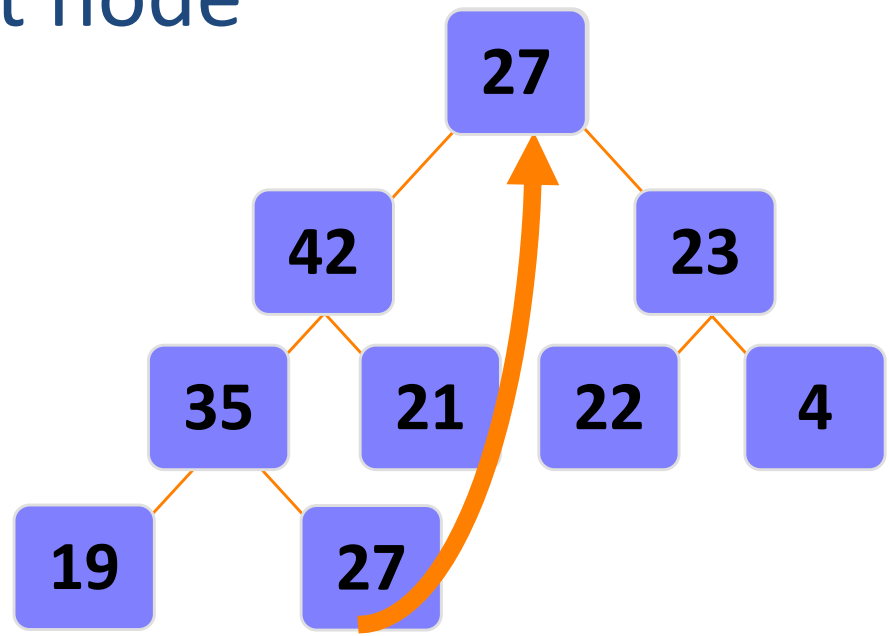
Operation - **Deletion**

- It is similar to STACK deletion
- Only Value at ROOT will be deleted
- Fill the space by last node
- Move downwards
  - Percolate Down



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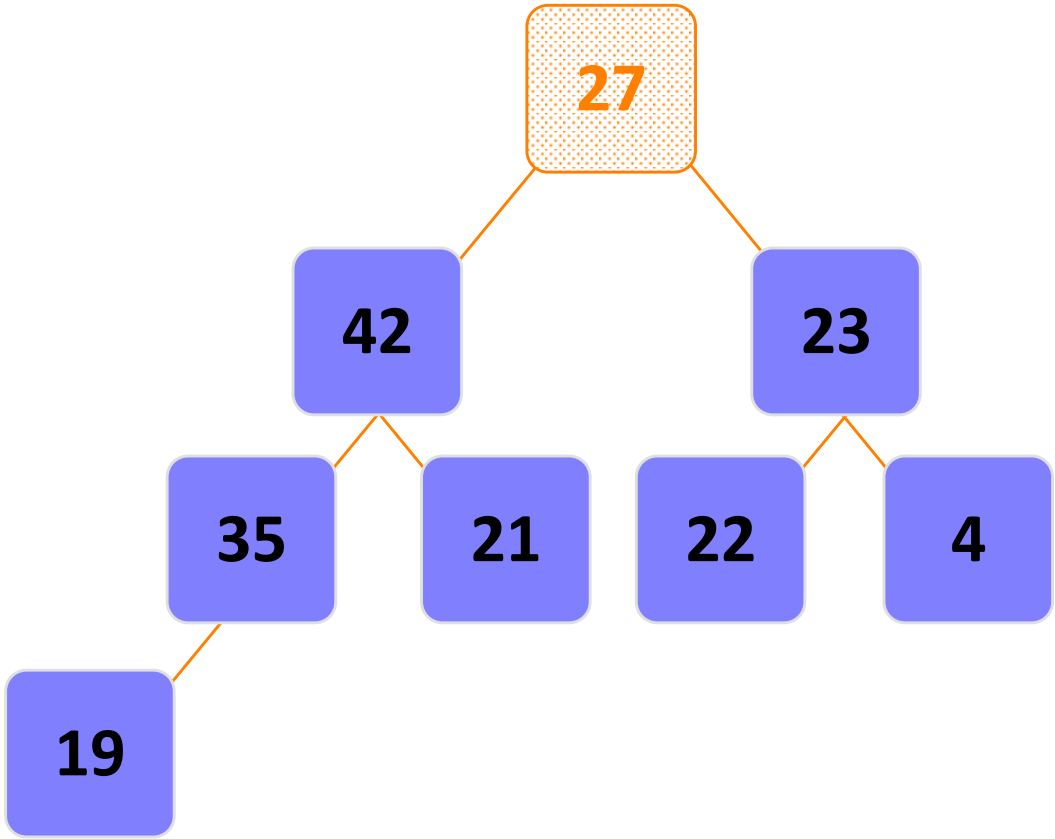
Deletion – Percolate Down

- If new node < children node(s)
  - Swap them
- Continue till
  - new node > Children nodes OR
  - New node become "leaf"



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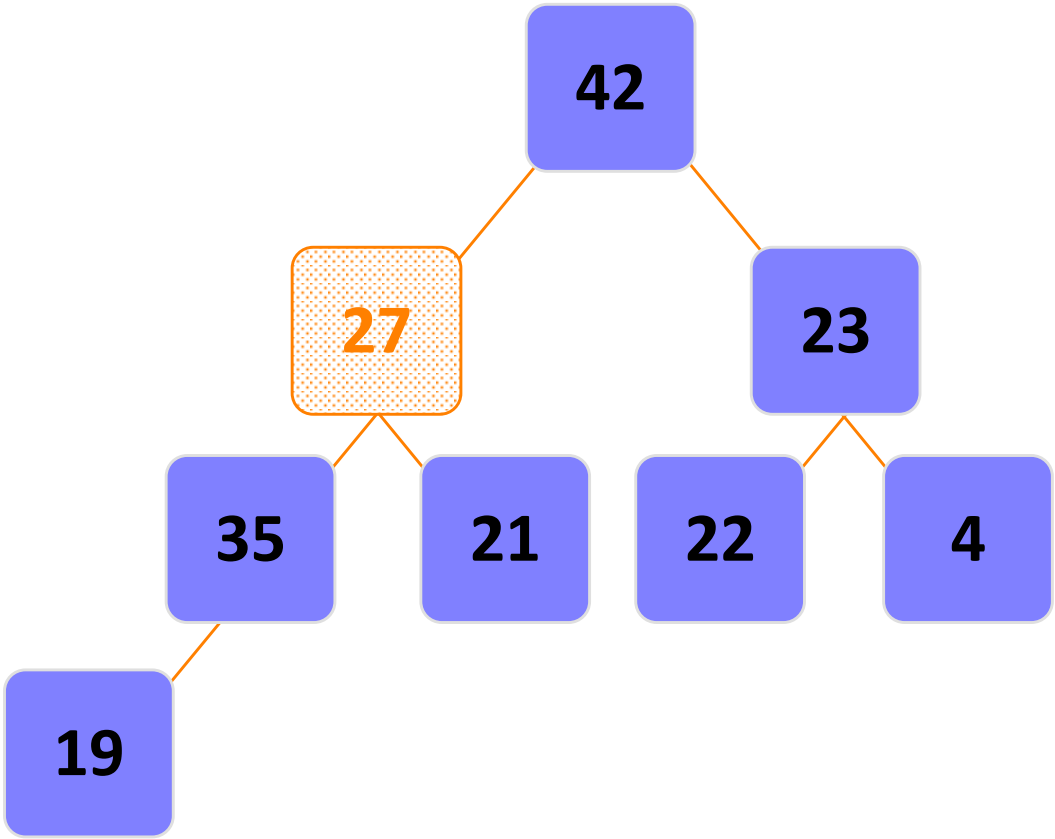

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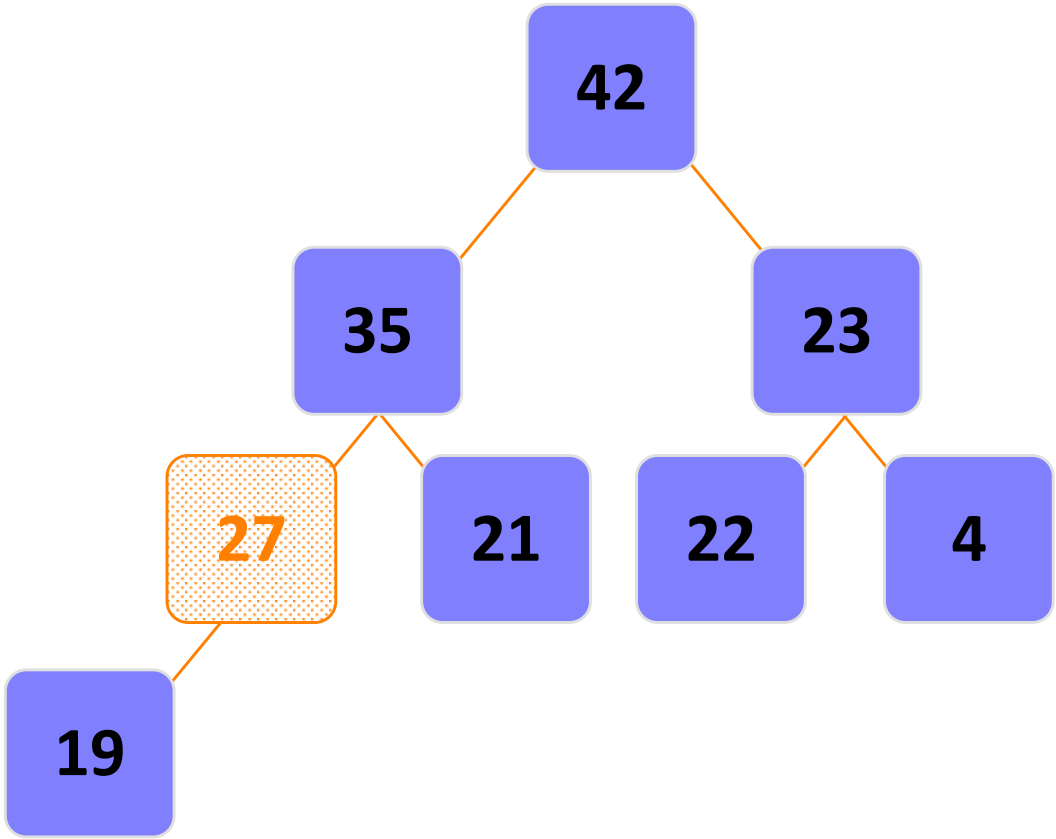

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Deletion – Percolate Down

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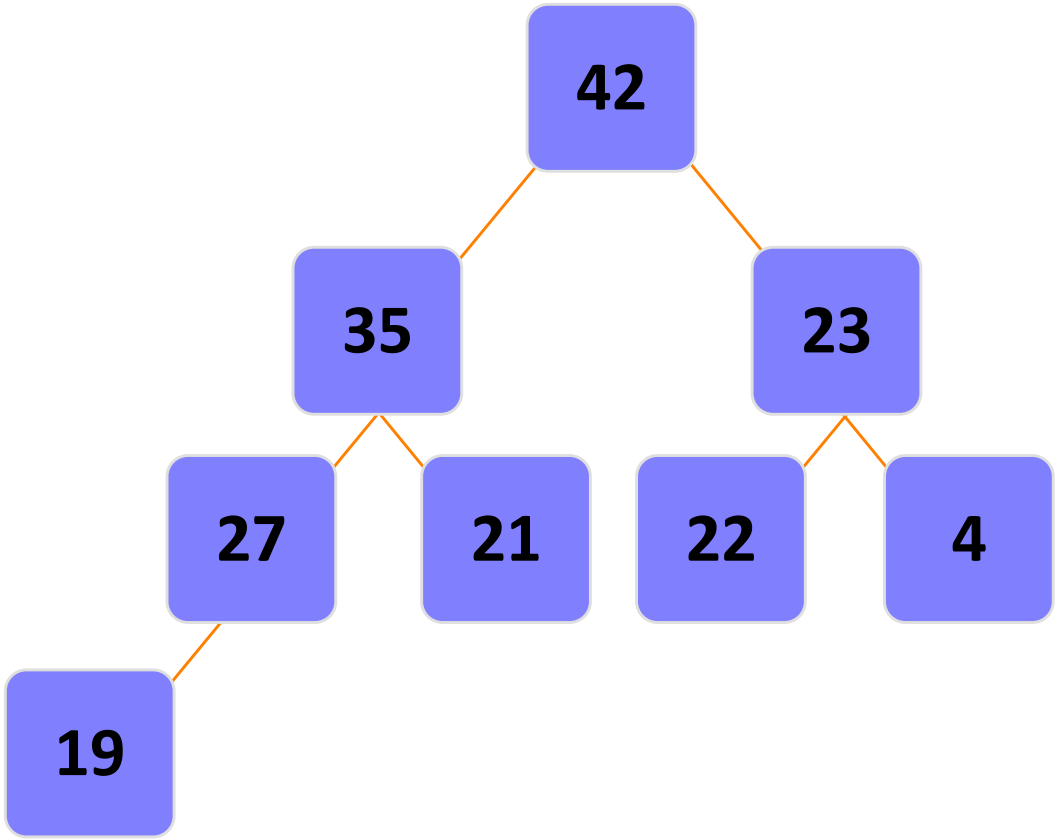

Deletion – **Analysis**

- What is the Big-O running time of Deletion from a heap?
- The actual deletion from root
  - O(1)
- Percolate Down
  - Height of tree
  - O(logn)
- Overall **O(logn)**

CPCS 204

Heap Operations

Creation

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CPCS 204

Heap – Creation

- Heap is Empty
- First value will make the "root"
- It is really just a series of "insertions"
- Simply insert the n elements into the heap in the order that they arrive (in our case, from left to right)
- Heap is created

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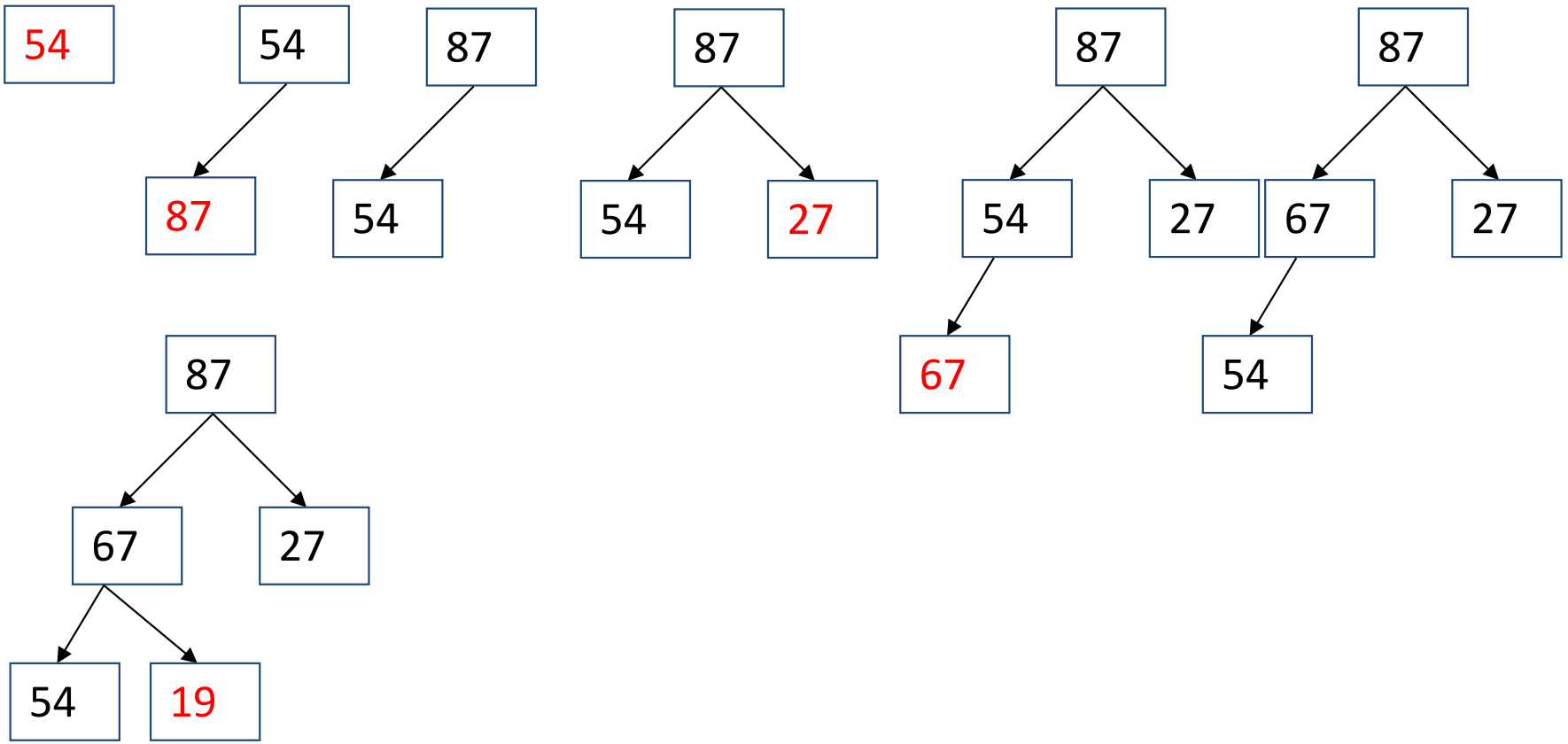
Heap – Creation

- 54, 87, 27, 67, 19, 31, 29, 18, 32, 56, 7, 12, 31



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Heap – Creation

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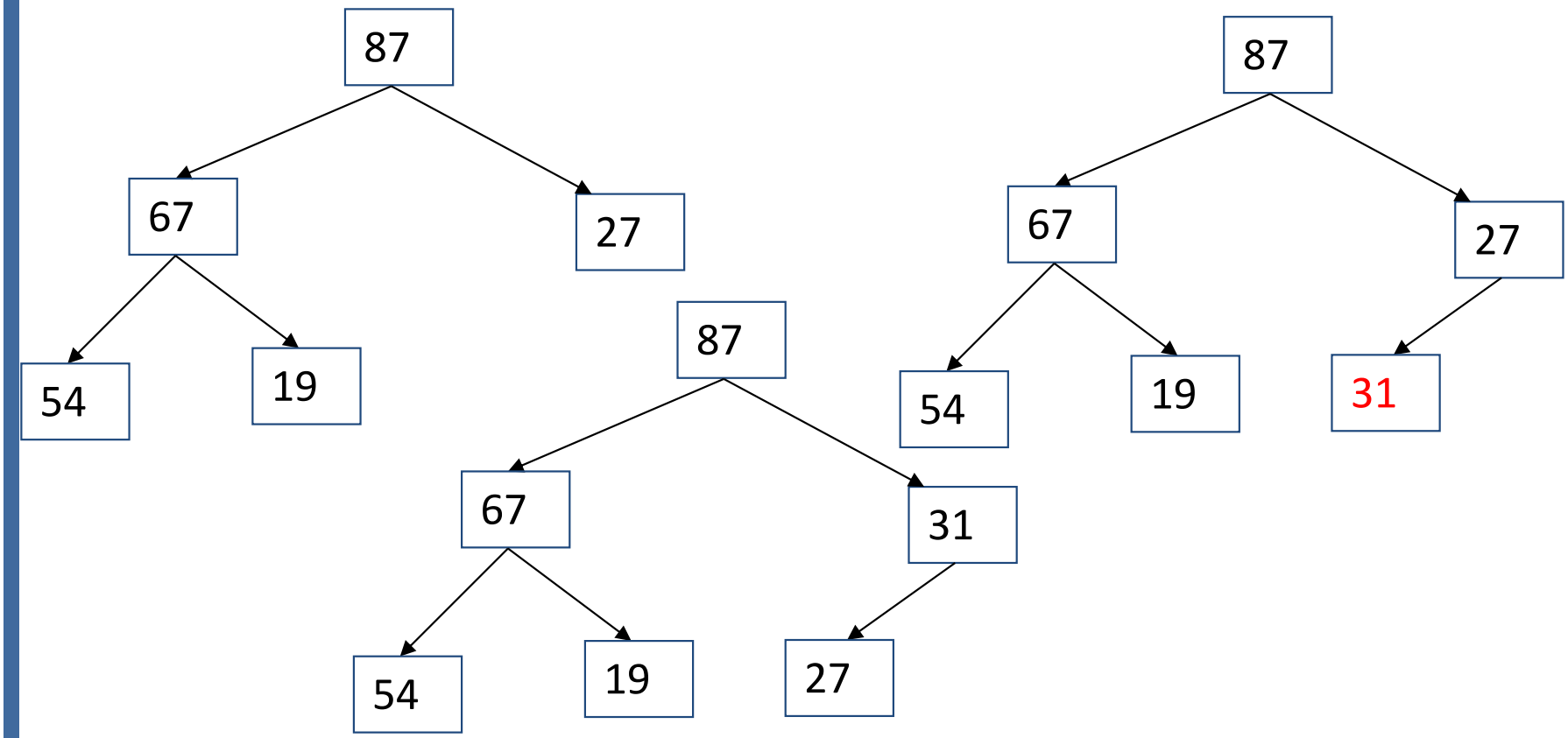

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Heap – Creation

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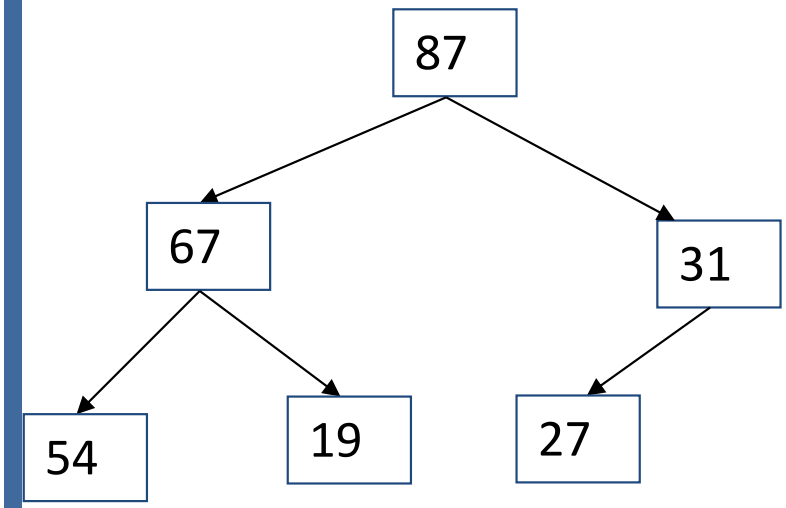

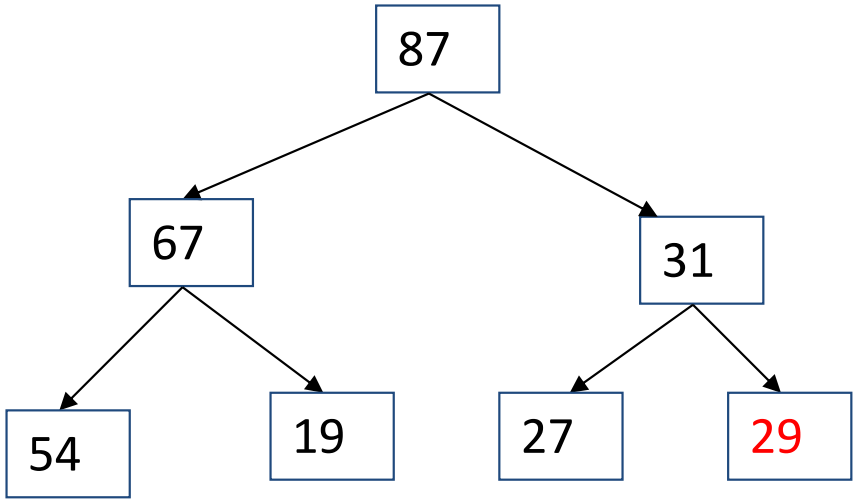

Heap – Creation

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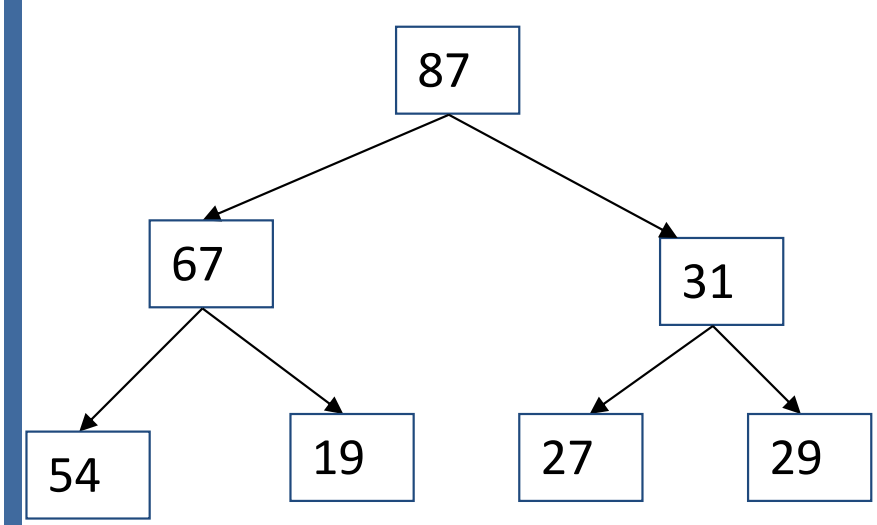

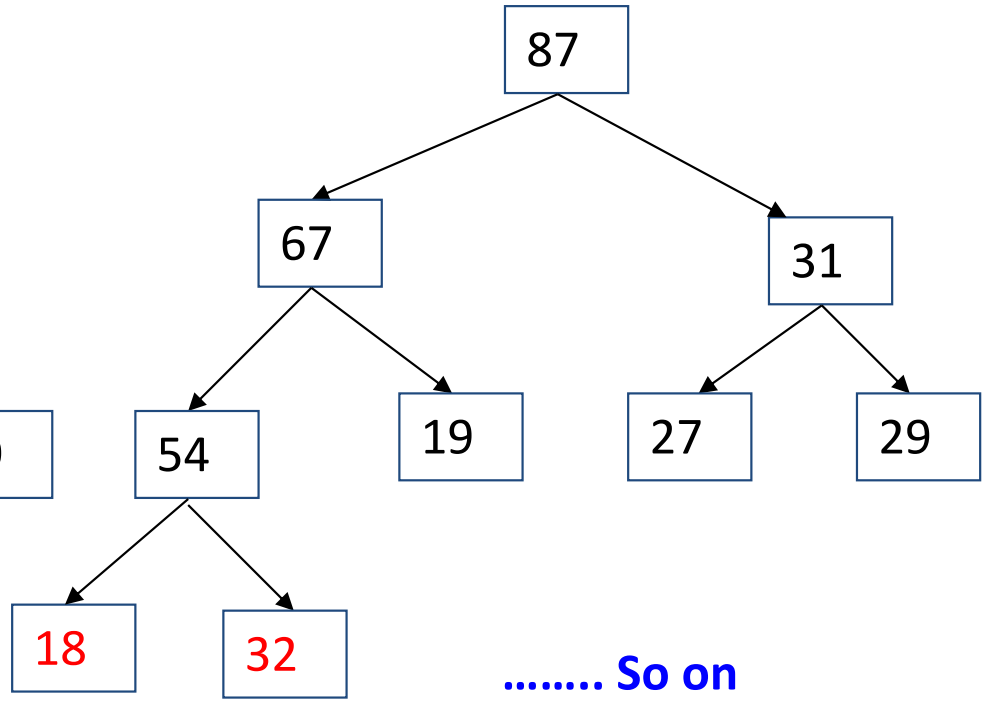

..... So on

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Creation – **Analysis**

- What is the Big-O running time of building a heap?
- Inserting "one" element
  - O(log n)
- Inserting "n" elements
  - O(nlogn)
- Building heap **O(nlogn)**

Data Structures - I

"No Bonus" marks for this course anymore

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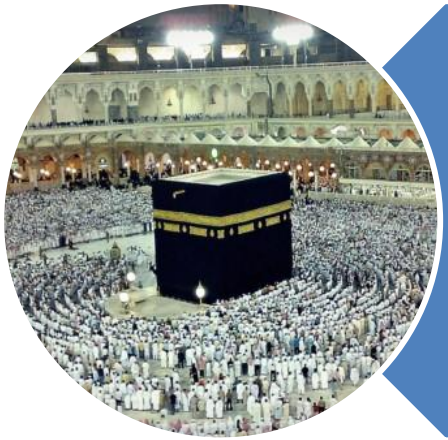

Allah (Alone) is Sufficient for us, and He is the Best Disposer of affairs (for us).(Quran 3 : 173)

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